GEK-83859

DRIVE SYSTEM INSTRUCTIONS

POWER CONVERSION MODULE
DS3820PCD
(FORMS E & G)

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

POWER CONVERSION MODULE

DS3820PCD

(FORMS "E" & "G")

(GEK - 83859)

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NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, 1501 Roanoke Boulevard, Salem, Virginia, USA, 24153.

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SAFETY SYMBOL LEGEND



Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.



Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to, or destruction of equipment.

NOTE

Commands attention to an essential operating or maintenance procedure, condition or statement which must be highlighted.

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FUNCTION

The power conversion module converts a constant frequency AC voltage into a variable DC voltage that can be used by a DC motor. The amount of power that the power conversion module delivers to the motor is controlled by the Control Electronics Module.

OPERATION

The power conversion module consists of three major parts:

- The SCR Bridge
- Cooling System
- Control of High Voltage Interface

The SCR Bridge contains the SCRs (Silicon Controlled Rectifiers) which will conduct current by precisely controlled gating signals. Thus a variable DC voltage is produced, which controls the motor speed. In addition, the Bridge contains anode to cathode snubbers, and line/leg reactors. The snubbers limit the rate of rise of voltage and the reactors limit the rate of rise of current to within the design limit of the SCR.

Heat generated by the SCR bridge during operation is removed via the heat sinks. A single cooling fan mounted on top of the PCM Bridge pulls air up over the heat sinks to remove the excessive heat. The Control Electronics Module monitors the cooling fan via a fan mounted centrifugal switch (closed = fan operating). 28VDC, originating at the Control Electronics Module is routed through the fan centrifugal switch, returning as "FANOK". If the fan is operating, the "Fan OK" circuit is completed, and is a permissive to operate the drive. Loss of "Fan OK" is a drive inhibit.

The High Voltage Interface consists of a printed circuit card. Its purpose is to scale down high voltage and current to low level signals for use in the control electronics. In addition, temperature sensors are mounted on this card to monitor the ambient temperature of the bridge and send a signal proportional to temperature to the Control Electronics Module.

RATING

Forms "E" & "G" of the DS3820PCD are rated 1360 Amps utilizing a 600VAC, three phase (3Ø), 3 wire, 60Hz input. The forms differ only in the location or connection points of the incoming AC power to the disconnect device. For example, Form "E" has front connected AC power, and Form "G" has rear connected AC power.

REMOVAL AND REPLACEMENT PROCEDURES

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

Removal and Replacement of PCM Cell Stack and Fuses

REMOVAL:

1. Remove all power from the panel (trip open PCP Switch Circuit Breaker). Open the panel door and then open the power conversion module door. Remove the two (2) screws that hold the clear Lexan® barrier on the front of the SCR cell stack. Pull the Lexan barrier from the slotted retainers.

2. Remove the ribbon cable clamp from the lower left hand sidewall. (All models do not have this clamp.)

3. Remove four sets of bolts, washers, and lockwashers retaining the two fuses and remove the fuses. Lay spacers with wires attached to one side.

   NOTE: If only fuses require replacement, proceed to REPLACEMENT procedure at this point.

4. Gripping the "ears" of the Mate-n-Lock plug, squeeze inward and pull upward on the plug at the top and downward on the plug at the bottom to disconnect the module harness from the gate leads.

5. Remove the nut, washer, and lockwasher holding the bottom of the stack assembly.

6. While holding the stack assembly in place by hand, remove the bolt, washer, and lockwasher holding the top of the stack assembly. This releases the stack assembly for removal.

7. Move the stack assembly toward the front of the module carefully, being aware of the "stud" in the bottom hole. Remove the assembly from the module.

REPLACEMENT:

1. Clean mating surfaces of the replacement heat sink with #360 or finer sandpaper. Apply #2 EJC grease or equivalent, wipe clean, and apply a fresh coat.

2. Reassemble in reverse order of the removal procedure described above. Torque stack assembly and fuse mounting hardware to 12.5 foot-pounds. Lexan baffles must be replaced exactly as they were to prevent possible overheating of the cells.

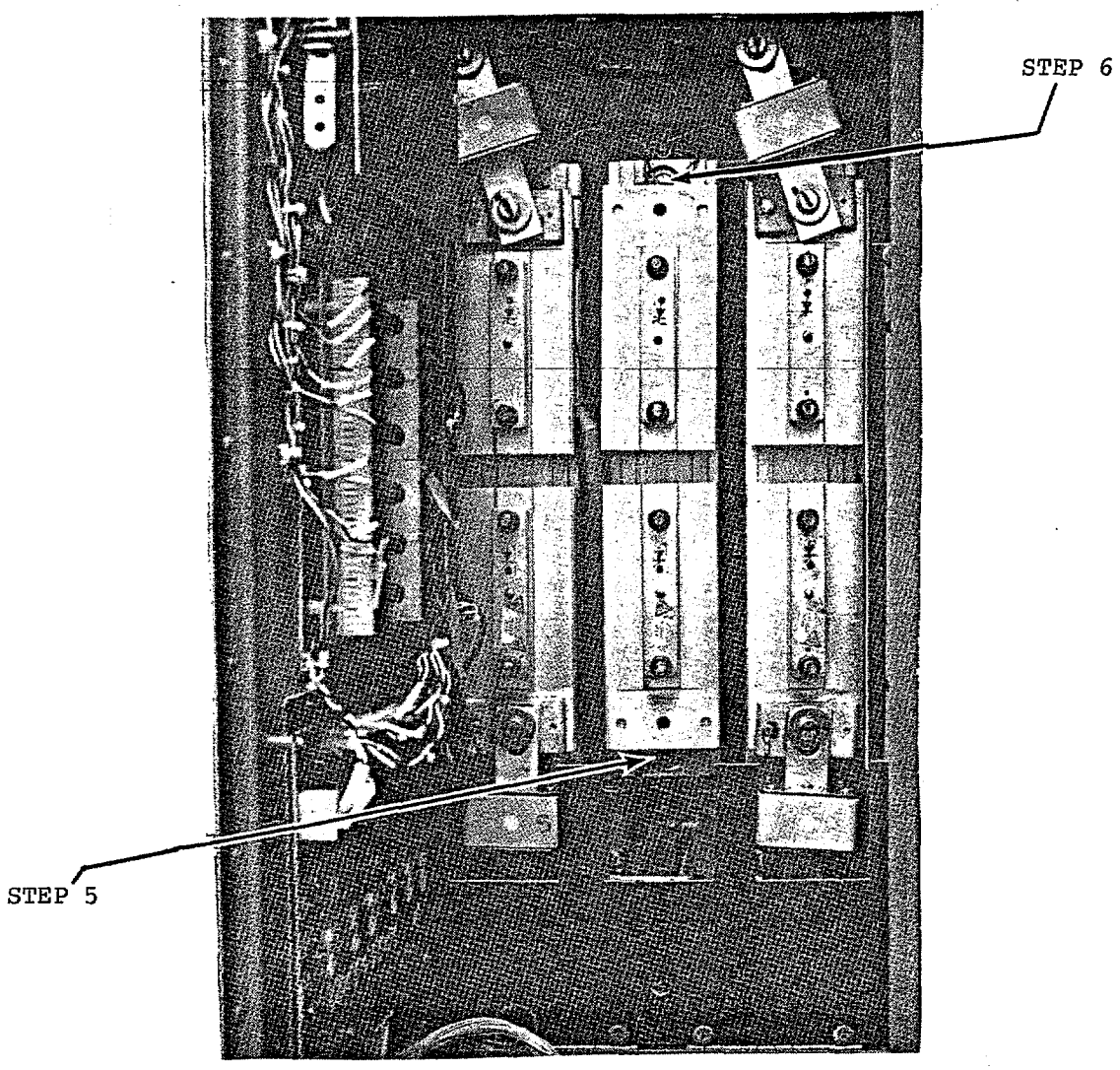


Figure 1 - PCM CELL STACK ASSEMBLY READY TO BE REMOVED (FUSES ALREADY REMOVED)

REPLACEMENT (Continued):

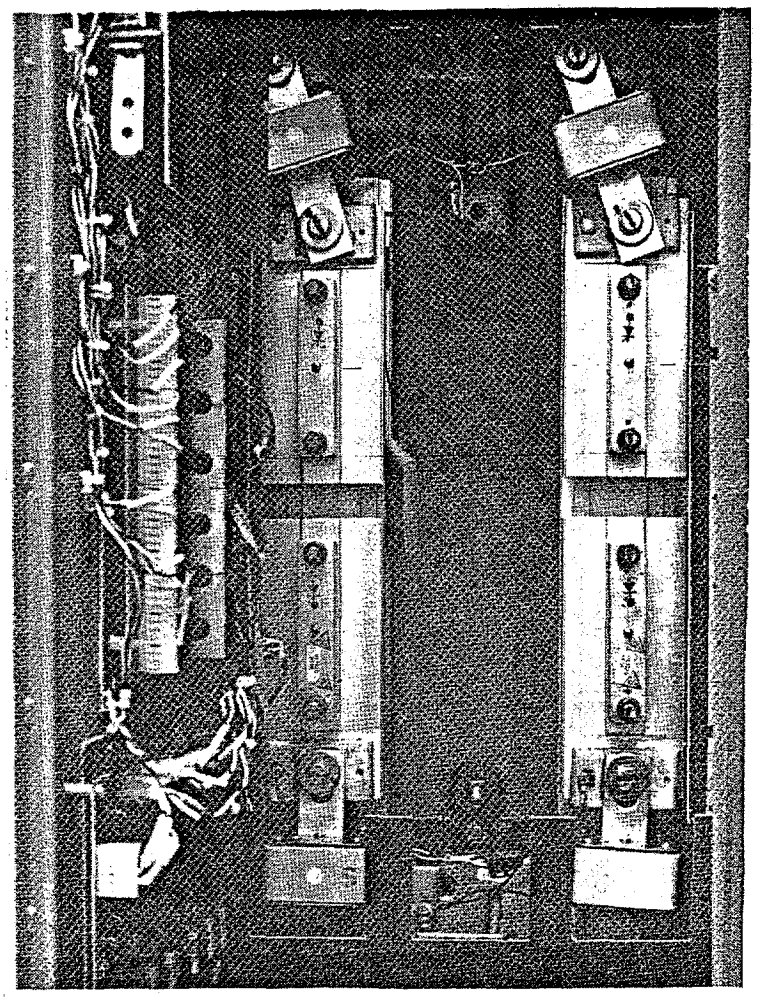


FIGURE 2 - PCM WITH ONE CELL STACK ASSEMBLY REMOVED

Removal and Replacement of PCM Fan

REMOVAL:

NOTE: Prior to fan removal, check fan power circuit for blown fuses (overload caused by single phasing three phase circuit), loose Fan OK circuit, or other problems external to the fan.

1. Remove all power from the panel and remove any devices obstructing fan removal. Open the PCM door and remove six (6) screws from fan plenum chamber, located on top of the PCM module enclosure. Retainer nuts are captive and should not fall into the SCR stack. Unplug the centrifugal fan switch snap-lock connectors.

   Remove the fan three phase (3Ø) power connections (3 wires) connected to the three leg fuse block.

2. Grasp the fan assembly and firmly pull straight out (weight is about 40 pounds) and toward the front of the enclosure. Remove the fan assembly and place it on a table.

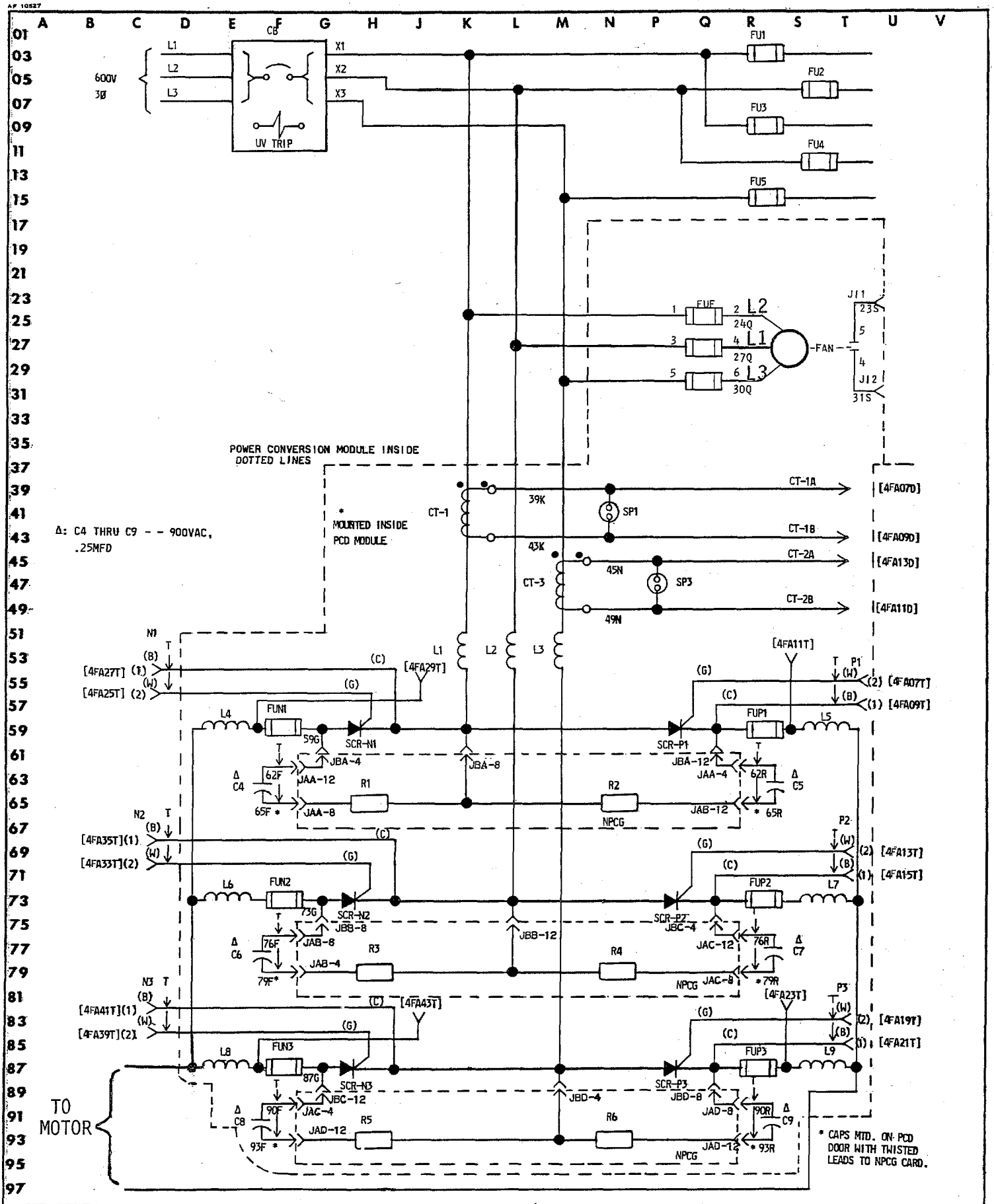
   Inspect the fan for obvious faulty conditions.

REPLACEMENT:

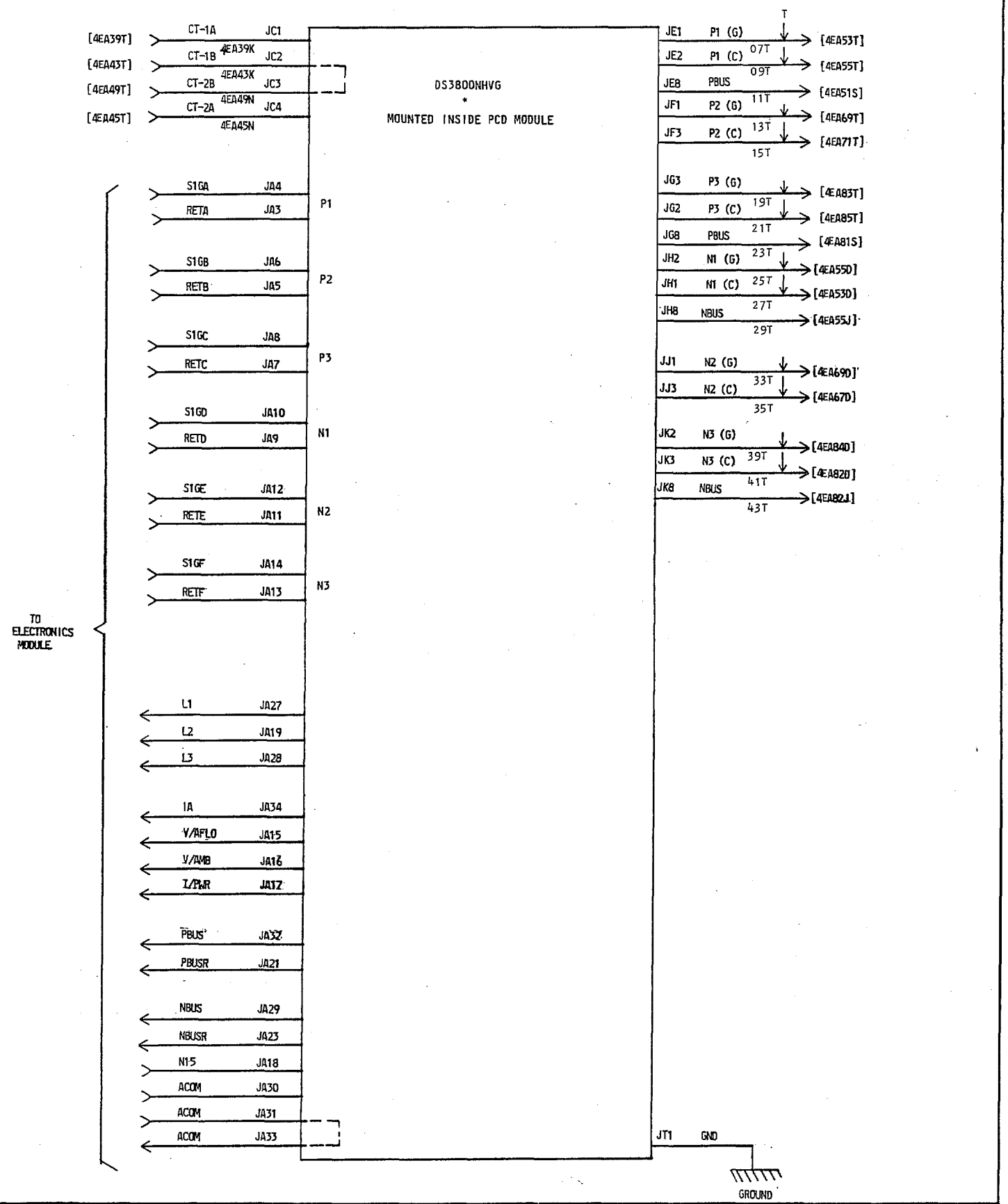
Replace the fan by reversing the order of the removal procedure you used.

ELECTRICAL DIAGRAMS:

The remainder of this book contains the electrical diagrams of the power conversion panel.



REV. 1	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM OIL WELL DRILLING POWER BRIDGE FORMS "E" AND "G"
REV. 2	ISSUED May 2, 1985					DS3820PCD
REV. 3	MADE BY F. BARTLEY			- 6 -		CONT. ON SH. 4FA SH. NO. 4EA



REV. 1	REV. 4	REV. 5	PRINTS TO OL114	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM OIL WELL DRILLING HIGH VOLTAGE BOARD FORMS "E" & "G"
REV. 2.	ISSUED May 2, 1985			- 7 -		DS3820PCD
REV. 3	MADE BY F. BARTLEY					CONT. ON SH. SH. NO. FA

FOLD

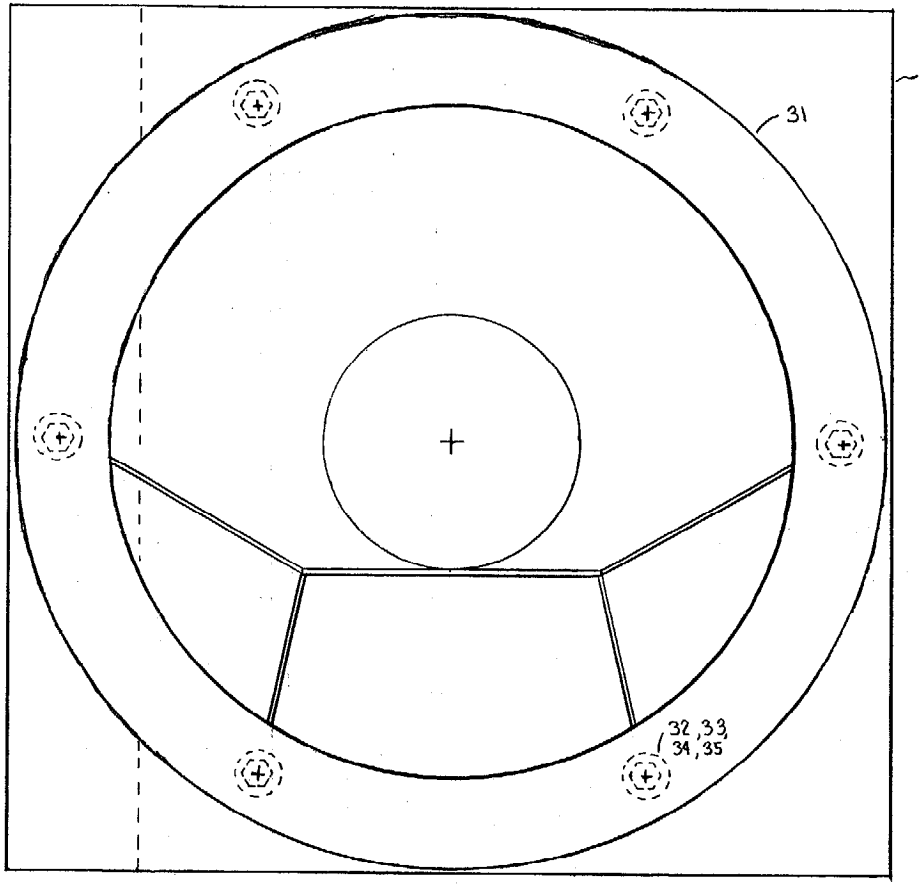
GENERAL ELECTRIC

DS3820FND CONT ON SHEET 6AD SH NO. 6AC

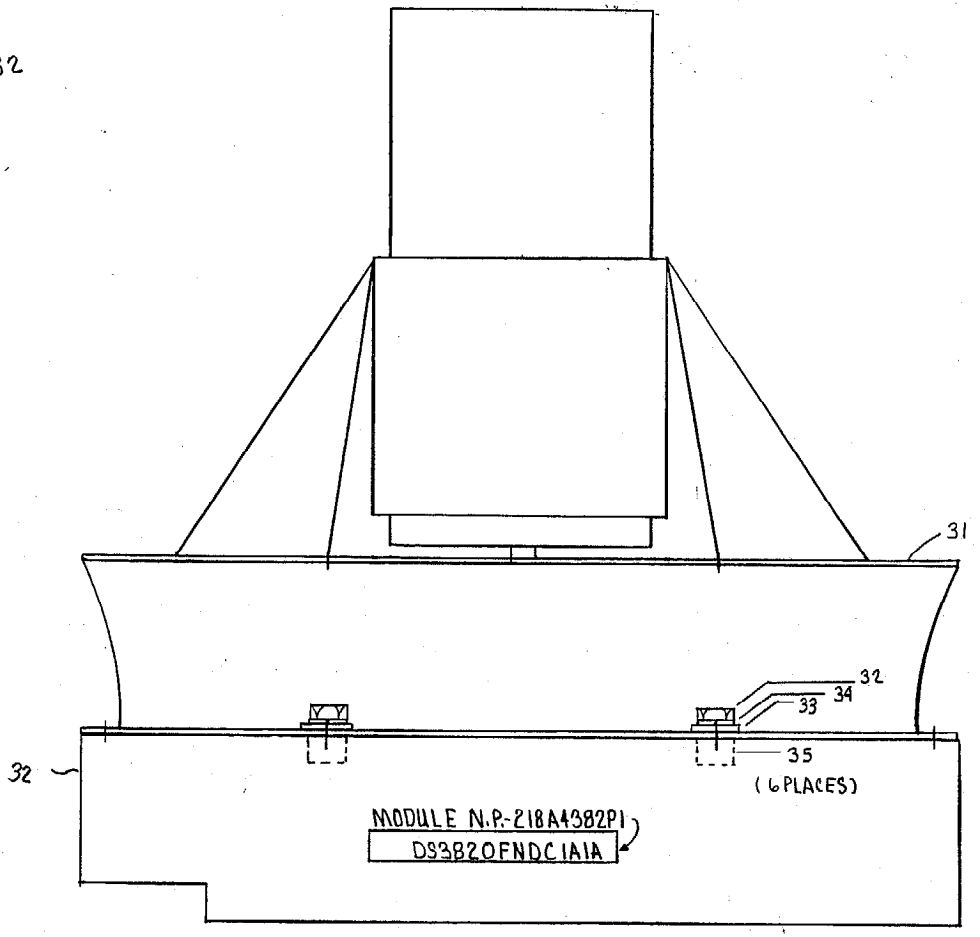
REV NO. DS3820FND CONT ON SHEET 6AD SH NO. 6AC

TITLE: FAN ASSEMBLY - FORMC

FIRST MADE FOR REQ. S.O.



TOP VIEW



FRONT VIEW

AIR FLOW

DL116

PRINTS TO

REVISIONS

MADE BY P.S. PATE ISSUED 12-7-84

APPROVALS P.B.P.

DRIVE SYSTEMS DIV OR DEPT SALEM, VA. LOCATION

DS3820FND CONT ON SHEET 6AD SH NO. 6AC

FOLD

FF-806WF (5-76) PRINTED IN U.S.A.

CARS

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GEK - 83859 POWER CONVERSION MODULE - DS3820PCD (FORMS "E" & "G")

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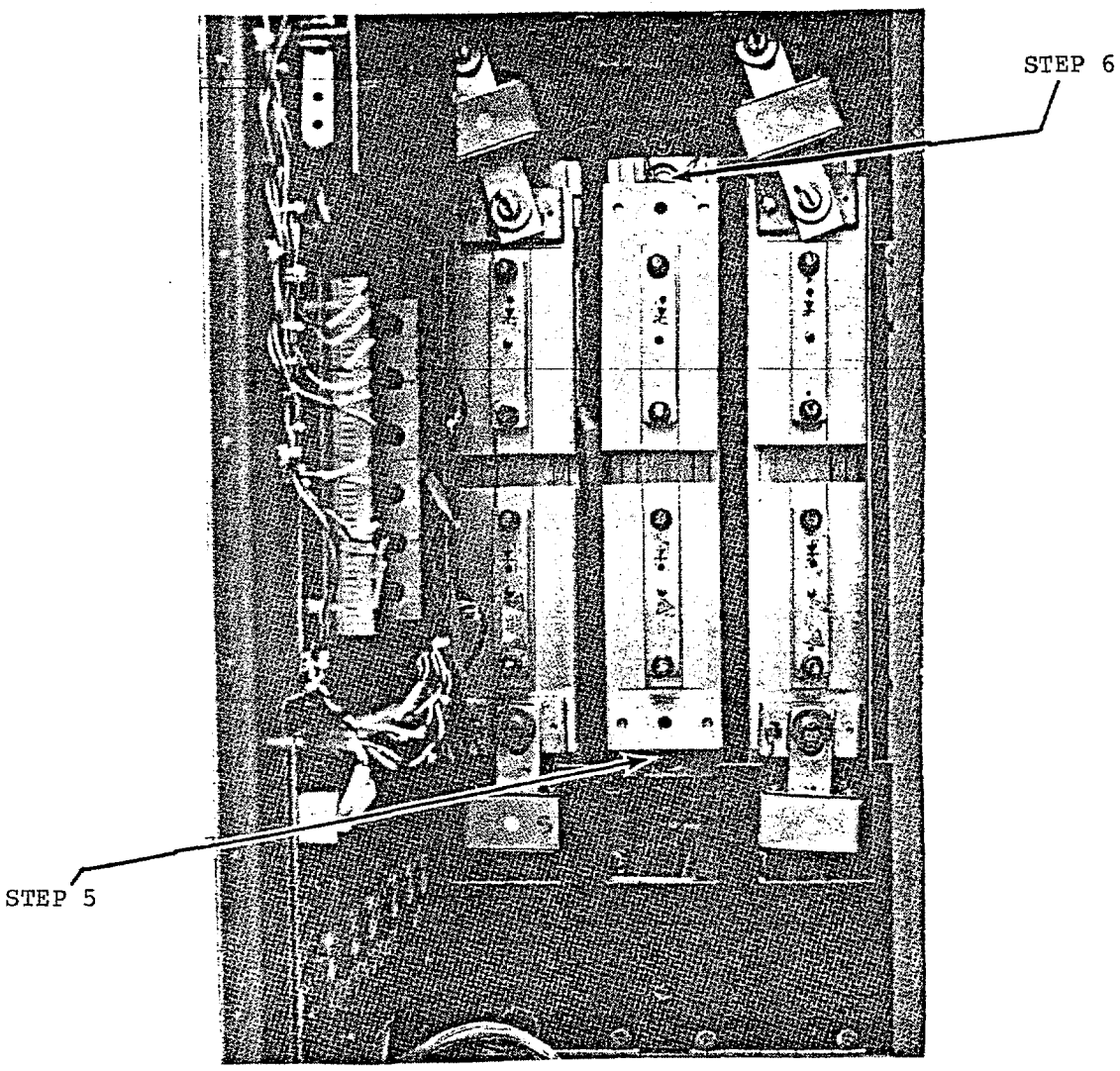


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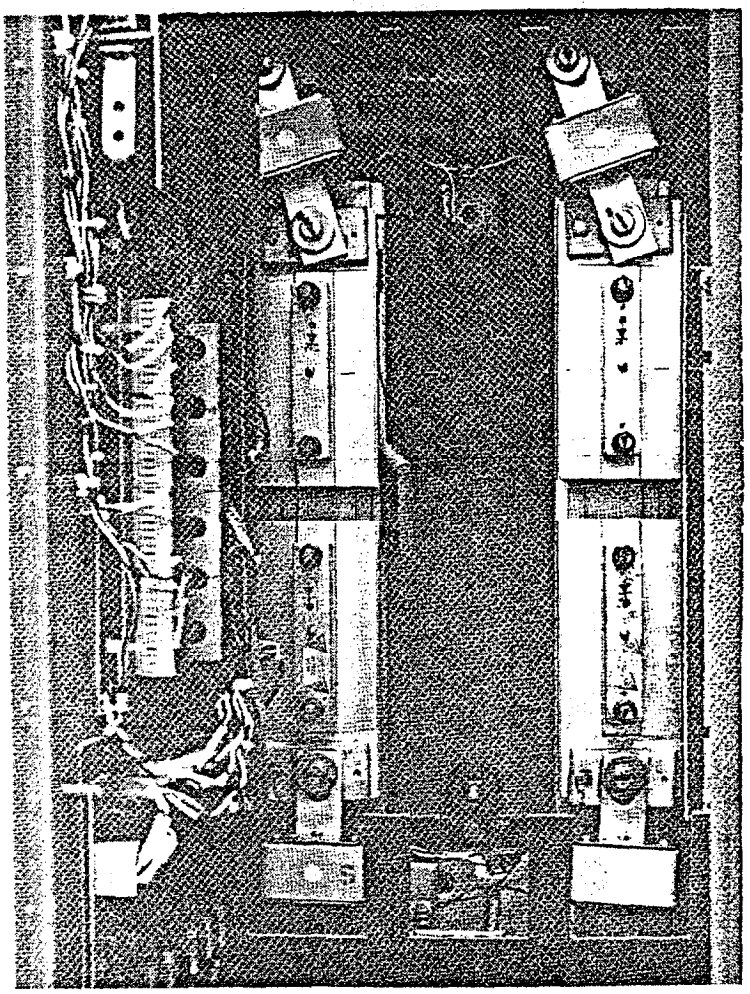


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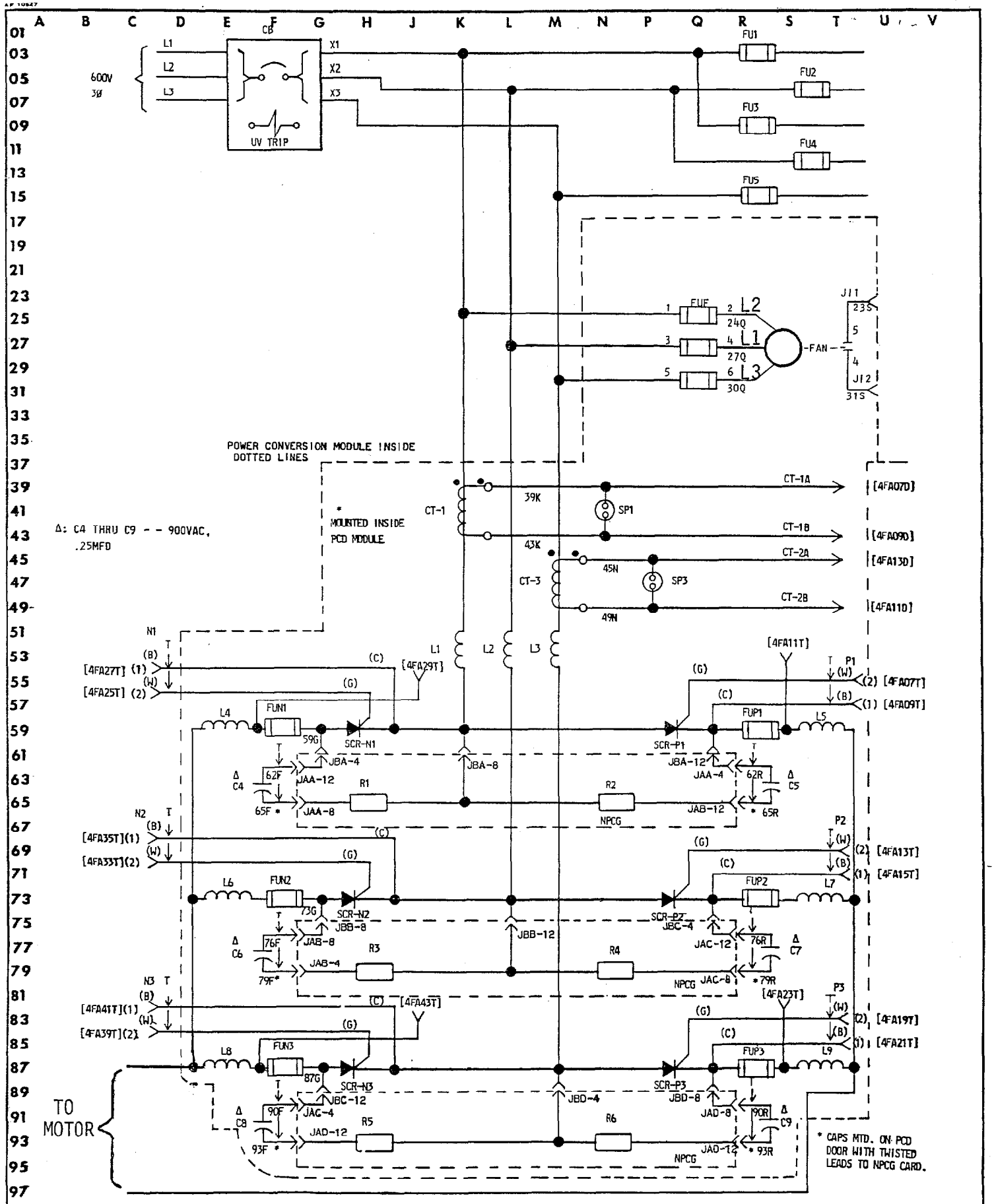
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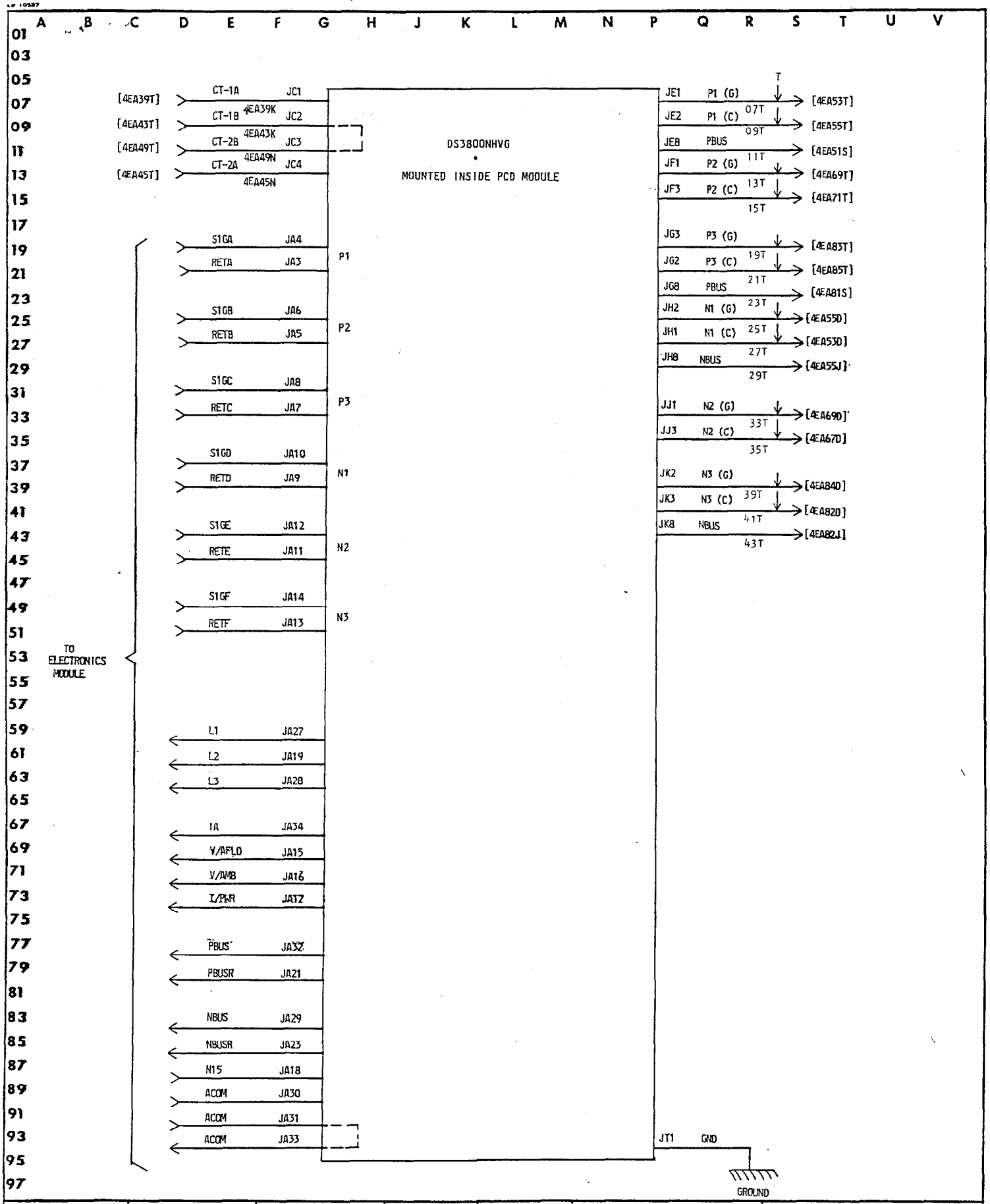
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REV. 1	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED May 2, 1985				DRIVE SYSTEMS DEPARTMENT	OIL WELL DRILLING POWER BRIDGE FORMS "E" AND "G"
REV. 3	MADE BY F. BARTLEY			- 6 -	SALEM, VA. U.S.A.	DS3820PCD
						CONT. ON SH. 4FA SH. NO. 4EA



REV. 1	REV. 4	REV. 5	PRINTS TO	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
			DL114		DRIVE SYSTEMS DEPARTMENT	OIL WELL DRILLING
REV. 2	ISSUED					HIGH VOLTAGE BOARD FORMS "E" & "G"
	May 2, 1985				SALEM, VA. U.S.A.	DS3820PCD
REV. 3	MADE BY			- 7 -		CONT. ON SH. SH. NO 4FA
	L.L.	F. BARTLEY	8144			

GENERAL ELECTRIC

D9382OFND
CONT ON SHEET 6AD SH NO. 6AC

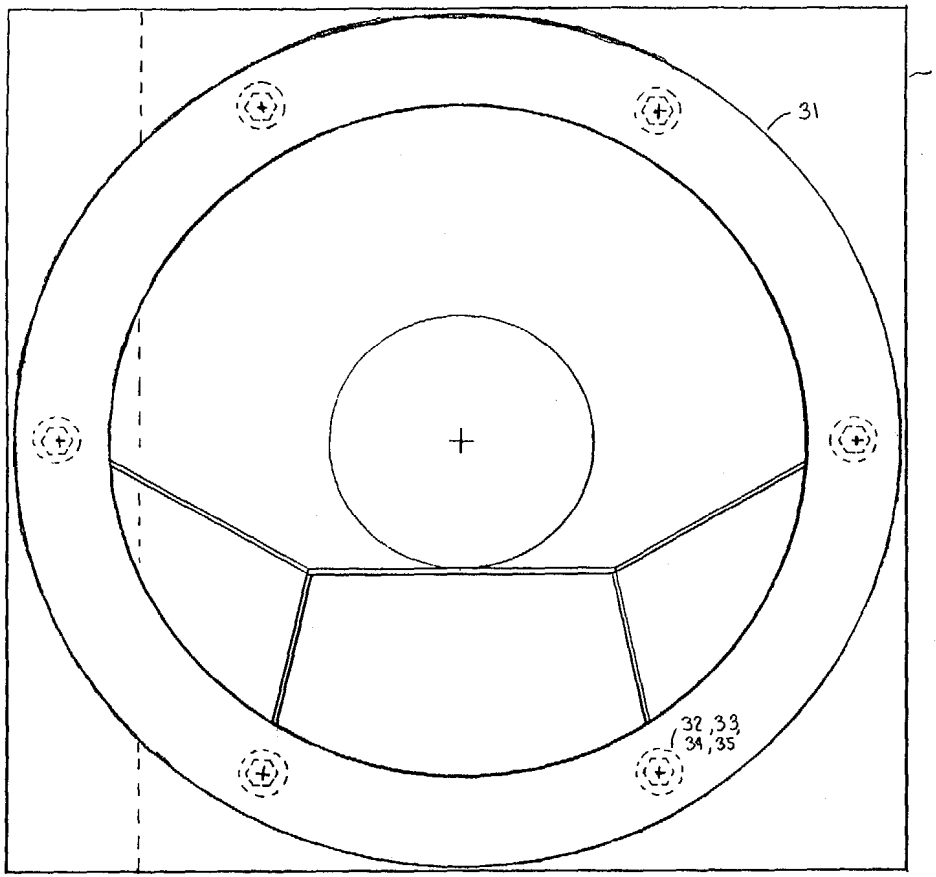
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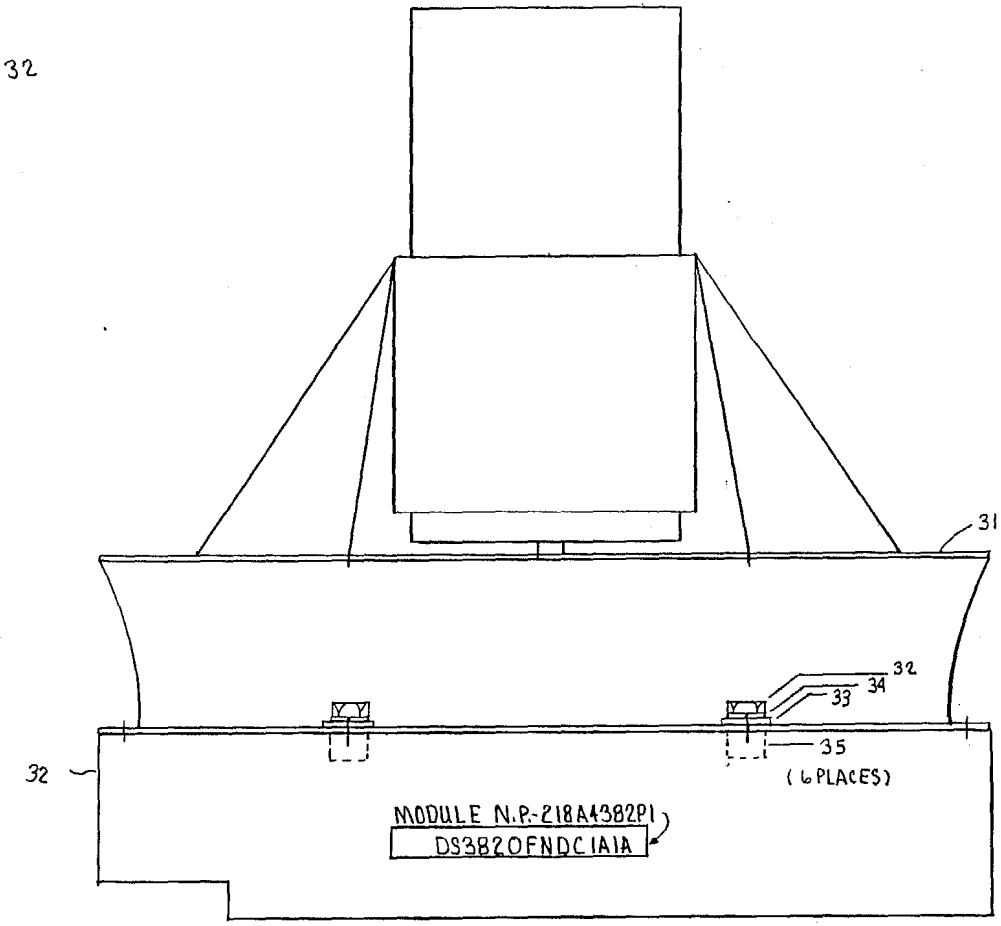
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CONT ON SHEET 6AD SH NO. 6AC

TITLE.

FAN ASSEMBLY - FORM C

FIRST MADE FOR REQ. S.O.





TOP VIEW

FRONT VIEW

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DL116

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GENERAL ELECTRIC